

The Young Athlete

Benton E. Heyworth, Mininder S. Kocher

Sports injuries in pediatric and adolescent athletes are being seen with increasing frequency, due to a variety of factors, including increased participation in higher levels of intensity and competition at younger ages, increased recognition of injuries in this age group, and the advent of arthroscopy and magnetic resonance imaging (MRI). The pediatric athlete differs from the adult athlete in terms of physiology, growth, psychology, and skills. Injury patterns are specific to the age of the athlete and the sport he or she plays. An understanding of the special considerations of pediatric athletes and common injury patterns in this population is necessary for the successful management of their sports injuries.

EPIDEMIOLOGY

Pediatric Sports Participation

During the past 30 years, the number of children and adolescents participating in physical activity and team sports has significantly increased, with the largest increase among female adolescents.¹ The overall trend has shifted from the largely unstructured, unsupervised “free play” of the early 20th century to the evolution of organized and highly structured youth sports activities.² It is estimated that currently up to 30 million children and adolescents participate in an organized sport in the United States. In 1995, reports indicated that 15 million 5- to 14-year-olds played baseball in the United States.³

The Youth Risk Behavior Survey (YRBS) was a large population-based study performed throughout the 1990s that enabled accurate assessment of emerging trends in youth sports participation. Results from the 1997 survey indicated that 62% of US high school students participated in one or more sports teams, with the majority playing in a combination of both school and non-school teams.⁴

The YRBS study highlighted a number of significant demographic differences when results were compared for age, gender, and ethnicity. Although the number of girls participating in sports teams has increased fivefold during the past 30 years, a disparity continues to exist between genders according to the 1997 YRBS study.¹ Although almost 70% of male high school students participate in sports, only 53% of similarly aged female high school students exhibit the same level of sporting interest.^{1,4} This gender disparity was even more dramatic among ethnic minorities, with only 40% of Hispanic and African-American girls participating compared with 62% and 71% for boys, respectively.⁴

Furthermore, progression into adolescence was also associated with a reduction in the involvement of both boys and girls in vigorous sporting activities.^{1,4} In boys, participation in vigorous exercise (defined as activity causing shortness of breath, lasting at least 20 minutes, 3 days a week) reduced from 81% in grade 9 to only 67% by grade 12.¹ As expected, this trend was even more pronounced in girls, with 61% of female ninth graders participating in vigorous exercise compared with only 41% by 12th grade.¹

The growth and increasing popularity of school and community youth sports programs have become an integral part of American youth culture and have the potential to enhance the long-term physical and psychosocial health of children and adolescents who participate in these programs.⁴

Pediatric Sports Injury

Increased youth participation in sports and physical activities and specialization in particular sports (and positions) at an earlier age have resulted in an increase in sports-related injuries as a result of trauma and overuse.² The annual incidence of sports injuries within the United States is estimated to be approximately 3 million, with up to 70% of those injuries resulting from youth sports activities.³ High school athletics account for more than 2 million injuries annually, including 500,000 doctor visits and 30,000 hospitalizations.⁵ More than 3.5 million children younger than the age of 14 years are treated annually for sports injuries.⁵ The financial costs of managing these injuries in 1996 was well in excess of \$1 billion.³

Pediatric sports injuries are often unique, not only in terms of the underlying pathologic findings but also with regard to the challenges in managing these injuries. Many patients participate in several teams during a given season, the rest periods between seasons are short if not nonexistent, and the demand for sporting success from parents, schools, and sporting establishments is increasing.⁶

Pediatric sports injuries can be classified according to the age of the athlete, the type of injury, and the type of sport/activity being played when the injury occurred.⁷ From an epidemiologic standpoint, these classifications assist in identifying potential risk factors for injury and implementing prevention strategies and rehabilitation plans that are appropriate for the age of the patient and the sport being played.

Several studies have identified a correlation between an increased risk of sports-related injury and the increased age of

Abstract

Due to broader participation, greater intensity of training and competition, and trends towards sports specialization, youth sports injuries are becoming more common. Conditions once thought limited to adult athletes are being seen in the pediatric population, and adult principles of treatment must be modified in their applications to the different physiology, psychology, and growing musculoskeletal systems of the young athlete. Novel techniques, comprehensive approaches to care, a greater emphasis on sports injury prevention are all critical in stemming this rising tide of often preventable youth sports injuries.

Keywords

youth sports
young athlete
pediatric sports injuries
injury prevention
adolescent athletes

the pediatric athlete.⁷ A number of explanations for these findings have been postulated, including a greater opportunity for injury in the adolescent athlete because of longer game times, along with more frequent and intense practices.⁷ The provision of medical assistance at many high school and college games allows increased reporting of injuries.⁷ It appears that anatomic factors, such as the increased size of the athletes and the resultant increased force and speed of collisions, play an insignificant role, because the same trend was noted for both contact and noncontact sports.⁷

Sports injuries can be broadly divided into acute traumatic and overuse-type injuries according to their pathophysiologic characteristics.⁷ Although many acute traumatic injuries are the result of random events, overuse injuries are often the result of entrenched training errors and therefore have greater potential for prevention.⁷ The difficulty lies in identifying these overuse injuries because initially they can be only subtly disabling when compared with an immediate fall to the ground after a sprain, for example.

It is important that an injury be viewed in the context of the sport in which it occurred because an injury that may be functionally disabling for one sport may have no relevance in another sport.⁷ Furthermore, it is important for physicians to recognize that time lost from sports participation is often more of a concern to athletes, their families, and their coaches than the nature of the injury itself.⁷ These perceived differences in injury severity inevitably affect management programs.

Among school athletes, football has the highest rate of injury, with wrestling not far behind.³ The rate of injury in both males and females at the high school and college levels is comparable, with the exception of knee injuries, which occur at a slightly greater rate in females at the college level.⁸ Fortunately, fatal sports injuries are rare. Mueller and Cantu⁹ reported 160 nontraumatic deaths in high school and college athletes in the United States between 1983 and 1993, with the primary cause being heart related; only a small number of the injuries were heat related. These investigators also reported 53 traumatic deaths from 1982 to 1992 in football players, resulting primarily from head and neck trauma.⁹

OVERUSE INJURIES AND SPORTS SPECIALIZATION

In the past several years, increased attention has been paid towards the role of sports specialization among youth athletes and the reported increase in pediatric overuse injuries. A number of overuse injuries that are unique to adolescents and preadolescents, such as apophysitis, apophyseal avulsion injuries, and physal stress injuries, have been reported in higher numbers.⁷⁴⁻⁷⁵ In response to such trends, the American Medical Society for Sports Medicine (AMSSM) published a position statement in 2014, in which the authors articulated modern definitions of overuse injuries, provided an evidence-based review of trends adversely affecting the musculoskeletal health of young athletes, and delineating the risk factors for pediatric sports injuries and the phenomenon of “burnout.”⁷⁶ Importantly, “sports specialization” was defined as intensive, year-round training in a single sport at the exclusion of other sports. Although a systematic review of the literature related to youth sports specialization in 2016

demonstrated that historical evidence for this phenomenon was scarce, studies are rapidly emerging elucidating its adverse effects.⁷⁷ Bell and McGuine, in a series of studies on high school athletes, demonstrated a clear association between specialization and increased injury risk in adolescents.⁷⁸⁻⁸⁰ Moreover, although specialization was shown to increase over the career of high school athletes who progressed to division 1 college athletics, early specialization was *not* shown to be a requirement for progressing to this higher level of athletic success.⁸¹ Myer et al. described the risk factors for injury in young athletes to be year-round single-sport training, participation in more competition, decreased age-appropriate play, and involvement in individual sports that require early development of technical skills.⁸²

Youth Sports Injury Prevention

With recent evidence more firmly establishing the association between injury and specialization, the focus of many authors has shifted towards identification of factors reducing injury rates.⁸³ For example, physicians, coaches, and parents have been targeted as critical figures in the encouragement of young athletes towards greater consideration of cross-training, engagement of multiple sports through the course of a given year, and longer periods of physical rest from specific activities. Not only will this provide for critical healing of tissues stressed by repetitive activities but also aids in the development of a greater diversity of motor skills. Periodic intervals of strength and conditioning, in place of active competition, and neuromuscular training programs are emphasized as important principles for injury prevention in the growing musculoskeletal system. For example, a number of different simple neuromuscular training programs, such as the FIFA-11 program an injury prevention program developed by a panel of international experts, with the support of the Federation Internationale de Football Association (FIFA), and other anterior cruciate ligament injury prevention programs, have been shown to decrease lower extremity injury rates and optimize performance.⁸⁴⁻⁸⁵ Continued study on the various factors that may decrease youth injury risks—from improved equipment to thoughtfully structured approaches to training—will be essential in keeping young athletes on the field and out of physicians’ offices. The STOP Sports Injuries program, a public outreach campaign sponsored by the American Orthopaedic Society for Sports Medicine (AOSSM) and designed to facilitate prevention of youth sports injuries, is emblematic of the larger transition from treatment towards avoidance of overuse-related conditions before they arise, many of which are highly preventable.⁸⁶

EXERCISE PHYSIOLOGY

Endurance Training

The increased popularity of endurance sports such as swimming, running, rowing, and cycling among children and adolescents has heightened awareness of aerobic training as a means of maximizing performance.^{10,11} The beneficial effect of aerobic training in adults is now well established, with increases in maximal oxygen uptake (VO_2max) of up to 15% to 20% reported in the literature.¹¹ However, the ability to enhance the aerobic capacity of children and adolescents through endurance training remains

controversial because many of the studies to date have been methodologically flawed and largely neglected adolescents.¹⁰⁻¹⁴

Although several physiologic parameters may be used to measure aerobic fitness, VO_2max is the most commonly used in studies involving adult endurance.^{10,11} The usefulness of this parameter in children has been questioned because most children fail to ever reach the plateau consistent with VO_2max .^{10,11} As a result, VO_2max has been replaced with peak VO_2 in pediatric endurance studies, which instead measures the highest VO_2 level achieved prior to the point of voluntary exhaustion.^{10,11}

Despite the traditional view that prepubescent children are incapable of improving their aerobic capacity through endurance training, evidence to the contrary is now emerging in the literature.^{10,11} A review of 22 studies by Baquet et al.¹⁰ demonstrated that a 5% to 6% increase in peak VO_2 among both children and adolescents is possible with appropriate aerobic training. The ability to achieve these increases is influenced by several factors, including baseline peak VO_2 levels, program design, maturity level, and genetics.^{10,11}

The role of pubertal status on a child's ability to enhance aerobic capacity through endurance training remains unclear because of a lack of quality longitudinal data.^{10,11} Early research indicated that for the same relative training intensity, greater gains in peak VO_2 were demonstrated for circumpubertal relative to prepubertal subjects.^{10,11} Two theories have been used to explain this finding: first, the existence of a so-called *maturational threshold* below which training-induced adaptations in aerobic fitness were physiologically limited, and second, the fact that the greater level of habitual activity among children maintained their VO_2 closer to its maximum potential, making additional increases in peak VO_2 more difficult to achieve.^{10,11} Although data are limited, evidence is slowly emerging to contradict these theories as a better understanding is gained of the role of genetic, environmental, and endocrine influences.^{10,11,14} High-quality longitudinal studies that document not only chronologic age but also maturity status are essential.¹¹

Designing a program that incorporates appropriate levels of training duration, frequency, and intensity is essential to achieving the desired increase in aerobic capacity.^{10,11,14} In their literature review, Baquet et al.¹⁰ found that three to four 30- to 60-minute sessions per week was optimal. Interestingly, no clear relationship was found between the length of training program and peak VO_2 improvement.¹⁰ Training intensity is generally defined in terms of the percentage of maximal heart rate.^{10,11} Several studies have confirmed that a heart rate that exceeds 80% of maximum is required to obtain significant increases in peak VO_2 .¹⁰

Comparison between continuous and interval training and the effect on peak VO_2 is limited to prepubertal children.^{10,14} Nine of the 16 studies reviewed by Baquet et al.¹⁰ demonstrated a significant increase in peak VO_2 after continuous training. However, only 3 of the 16 studies showed improvement when the heart rate was less than or equal to 80% of the maximum.¹⁰ The implementation of continuous training among children poses difficulties with regard to compliance and motivation.¹⁴ Interval training is not only easier to put into practice but has more consistently positive results. Programs that combine continuous and intermittent exercises make interpretation of results difficult.¹⁰

The increasingly competitive nature of sports has resulted in reluctance by athletes to take adequate breaks from training and performing.¹⁵ The damaging effects of prolonged endurance training on skeletal muscle and function are well documented in the literature, as is the huge capacity of human skeletal muscle for repair and adaptation, given adequate recovery time.¹⁵ A study by Grobler et al.¹⁵ demonstrated that, although minor exercise-induced muscle damage is a precursor for adaptation, the reparative capacity of skeletal muscle is limited and the cumulative effects of repetitive trauma and injury to skeletal muscle may lead to reduced performance, especially in long-distance runners. Further research is needed to investigate the limits of skeletal muscle regenerative capacity after a chronic injury.¹⁵

Flexibility

Extremes of joint and ligament laxity have important implications for the pediatric athlete because of the increased risk of both acute traumatic and overuse-type sporting injuries, in addition to a number of degenerative orthopaedic conditions, many of which have long-term implications for sports participation and performance.^{16,17}

Childhood is associated with a gradual reduction in flexibility, with the greatest loss occurring around puberty as a result of a growth-induced muscle-tendon imbalance.¹⁸ This loss of flexibility is less pronounced in females.¹⁸ Excessive tightness during this time of rapid growth is thought to play a major role in both acute and overuse-type injuries affecting in particular the lower back, pelvis, and knee.¹⁷ Slight improvements in flexibility are observed after the pubertal growth spurt in both males and females until early adulthood, at which point it plateaus and then starts to decline once again.¹⁸

Although only 4% to 7% of the general population meets all the criteria for generalized ligament laxity, evaluation of flexibility remains an essential component of the clinical assessment of a young athlete because it enables identification of the persons at increased risk, in addition to providing invaluable information for injury prevention and rehabilitation programs.^{17,19} Studies performed by Marshall and colleagues in 1980 demonstrated that increased flexibility was associated with a greater risk of sports-related injuries, particularly in sports requiring rapid change of direction or acceleration.¹⁷

Although several instrumented tests are available to test the flexibility of individual joints, simple screening tests such as the modified Marshall test devised in 1978 are more commonly used as a routine part of the clinical assessment of the young athlete.¹⁷ By measuring thumb to forearm apposition, the modified Marshall test can quickly identify extremes of flexibility that warrant further, more in-depth investigation and assessment that is relevant to the athlete's given sporting interest.¹⁷

Strength Training

Traditionally, strength training was discouraged among the pediatric population because of the perceived risk of growth disturbances and other injuries.¹⁶ However, research during the past 20 years has demonstrated that not only can strength training be a safe and effective component of any comprehensive fitness

program, but it can also provide clear health benefits to the pediatric age group.²⁰⁻²³ These benefits include improved athletic performance as a result of increased coordination, muscle strength, and power, in addition to enhancement of long-term health as a result of increased cardiorespiratory fitness, reduced risk of injury, and improved bone mineral density and blood lipid profile.^{21,22,24}

Research shows that expertly tailored strength-training programs in children and adolescents are associated with increased muscle strength and performance advantages in sports such as football and weightlifting.²⁴ Increases in strength of 50% to 65% above baseline have been reported in the prepubescent athlete over a 2- to 3-month training period.²⁵ However, in the preadolescent child, this increased strength occurs in the absence of muscle hypertrophy, highlighting the role of neurogenic adaptation as the likely cause. Neurogenic adaptation refers to the recruitment of increased motor neurons that can fire with each muscle contraction.²⁴ Moreover, the loss of benefits after the program is discontinued for 6 weeks provides further evidence for this hypothesis.²² In contrast, strength training during and after puberty is further enhanced by the hormonally induced increase in muscle growth that occurs in both males and females.²⁴

Although the risk of injury associated with strength training is real, research shows that it is no greater than in any other sport when adult supervision is available to ensure use of proper technique and implementation of safety precautions.^{22,24} Data obtained by the National Electronic Injury Surveillance System between 1991 and 1996 were used to estimate that strength training was responsible for more than 20,000 injuries annually in the group of athletes who were younger than 21 years.²⁶ However, the usefulness of these results is limited by the lack of distinction between competitive and recreational injuries or comments regarding the quality of the equipment being used or the presence of adult supervision.²⁴ Of note, 40% to 70% of the injuries were attributable to muscle strains, primarily within the lumbar area.²⁴ Case reports indicate that children and adolescents participating in strength training may be at risk of specific lumbar injuries, including herniated intervertebral disks, paraspinal muscle sprains, spondylolisthesis, and pars interarticularis stress fractures.²²

Thermoregulation and Heat-Related Injuries

Even though heat-related illnesses are preventable,²⁷ heat stroke remains the third most common cause of exercise-related death among high school athletes in the United States after head injuries and cardiac disorders.²⁸

Several physiologic characteristics unique to the pediatric population contribute to the thermoregulatory disadvantage they face in extreme climatic conditions, including increased surface area to body mass ratio, reduced sweating capacity, greater generation of metabolic heat per mass unit, and a slower rate of heat acclimatization.^{27,29} A large surface area to body mass ratio is advantageous in mild to moderate climates because of the increased convective surface it provides.²⁸ However, in hot, humid weather, this large ratio provides a larger area for heat influx, thereby raising the core temperature and increasing the risk of heat-induced illnesses.²⁸ Conversely, in cold climates, enhanced

metabolic heat production and cutaneous vasoconstriction are often insufficient to overcome the heat lost from their vast surface area, particularly in cold water.³⁰

Sweat glands play a central role in the pediatric athlete's ability to thermoregulate. By 3 years of age, the number of sweat glands a person shall possess is fixed.²⁸ Despite having a greater density of sweat glands per skin area than adults, the sweating capacity in children is restricted because of a lower sweating rate and a higher sweating threshold.³¹ As a result, the ability of children to dissipate body heat by evaporation is reduced until the transition is made to an adult sweating pattern in late puberty.^{27,29}

The reluctance of children to drink while engaging in prolonged exercise further exacerbates this thermoregulatory disadvantage.³² The American Academy of Pediatrics recommends prehydration, in addition to enforced periodic drinking during the course of prolonged exercise.^{27,33} Although water is readily available, flavored drinks are often easier for children to tolerate.²⁷ Moreover, because the risk of dehydration is even greater in children with certain diseases or conditions such as cystic fibrosis, diabetes, and anorexia, the need for optimal fluid intake during exercise is essential.²⁷

PSYCHOSOCIAL ASPECTS OF SPORTS PARTICIPATION

Psychosocial Development

Participation in sports activity is associated with a large number of health benefits that can influence both physical and psychosocial well-being. The social interaction associated with sports participation is instrumental in a child's psychosocial development, including character development, self-discipline, emotion control, cooperation, empathy, and leadership skills.³¹ The acquisition of new skills aids in building confidence and self-esteem.³¹ It also allows children to experiment with success and failure in a low-risk environment.³⁴

The YRBS study mentioned earlier in this chapter was a nationally representative study conducted throughout the 1990s by the Centers for Disease Control and Prevention. It evaluated the new trends in sports participation, with a particular focus on health behaviors.⁴ The study revealed a strong positive trend between sports participation and several types of positive health behaviors in white males and females, including consumption of fruit and vegetables as part of a healthy diet, reduced levels of smoking or illegal drug use, and a reduced risk of suicide.⁴ However, this trend was not found among ethnic minorities, and among Hispanics and African Americans, the risk of negative health behaviors actually increased with sports participation.⁴

Readiness for Sport

Knowledge of cognitive and motor developmental milestones and the factors that motivate children and adolescents to participate in sports is essential when designing sports activities that are both rewarding and beneficial.^{31,34} Motor development is a sequential process like any other developmental milestone, and the rate of progression varies among children.³⁵ Participation in most sports requires fundamental motor skills such as kicking, throwing, running, jumping, and catching.³⁵ Most children acquire these

skills through informal “play,” but mastery often requires more formal instruction and repetition.³⁵ Although this process of acquisition and mastery can potentially be accelerated through intensive instruction and practice, research shows that it rarely speeds up motor development or leads to enhanced athletic performance.³⁵

The principal motivating factors for young children to participate in sports activities are fun and enjoyment.³¹ For an activity to be viewed as enjoyable, it must include a certain level of excitement but ultimately a sense of personal achievement associated with the improvement or mastery of specific skills.³¹ We must acknowledge that, although virtually all children have the ability to acquire new motor skills, the ease of acquisition and degree of mastery may vary among children.³⁴ Research has shown that children who feel less competent with one particular skill are less likely to continue with that sport in the long term.³⁴ Therefore it is important that young children be exposed to a range of sports that challenge and enable them to acquire a variety of fundamental motor skills.³⁴

Progression into adolescence is not only associated with a number of physical changes resulting from the pubertal growth spurt but also a shift in the motivational factors influencing sports participation.^{31,34} Cognitive and motor development is now sufficient to allow for the incorporation of strategy into sports such as football or basketball.³⁵ The need for fun and excitement is overtaken by social factors such as interaction with friends and physical appearance, although mastery of skills remains important.^{31,35} Differing rates of progression through puberty can result in inequality within and between genders.³⁵ Persons who experience earlier growth spurts may be temporarily taller, heavier, and stronger, which often leads to unrealistic expectations because of the erroneous conclusion that they are destined to become better athletes than their less mature peers.³⁵

Adult Involvement

The level of adult involvement has increased significantly with the evolution of organized youth sports. Although the traditional role of “supervisor” still exists, the nature of adult involvement in youth sports has also evolved. An increased level of sophistication has developed as a result of the advent of specialized coaches, sports psychologists, nutritionists, and personal trainers, all of which undoubtedly affect the psychosocial development of the young athlete.

Adults are vital for the enforcement of rules and the creation of a safe, controlled environment in which to impart their knowledge and assist children and adolescents in the acquisition of new skills and the development of appropriate attitudes regarding sports.^{27,28} However, the involvement of adults in sports activities can also have a detrimental impact on psychosocial development through the expression of negative and unsportsmanlike behavior, negative reinforcement, and the enforcement of demands and expectations that exceed the child's abilities.^{31,34,37}

In the early years of life, parental influence is instrumental in the development of lifelong core values and attitudes.³¹ By 12 years of age, a child's attitudes to winning are already well established and often directly reflect the values held by their

parents.³¹ These values and attitudes are often acquired through observation of parental behavior, and although extreme parental behavior is rare, the use of negative comments or reinforcements is frequent.³¹ Variation was found between sporting codes, with the greatest incidence of extreme parental behavior occurring in soccer and rugby.³¹ Children of relaxed and supportive parents who positively reinforce their child's performance are not only more self-confident but are more likely to be successful athletes.^{31,34}

As the child progresses to adolescence, the role of parents starts to diminish as the role of the coach increases.³¹ Through the provision of feedback and reinforcement, coaches have a large impact on the confidence and self-perception of the young athlete.³¹

The increasingly competitive nature of sports has led to a shift in goals that are largely adult-oriented and focus on winning at any cost.³ Competitive behaviors start to emerge at 3 to 4 years of age, and the potential exists to either enhance or exploit this trait through the use of sports.³¹ The danger arises when the demands and expectations placed on young athletes by their parents or coaches exceed their abilities.³¹ This phenomenon can result in the development of unhealthy competitive behavior with serious antisocial interpersonal consequences or even problems such as burnout and chronic stress.^{31,36}

NUTRITION

The nutritional concerns of the pediatric athlete are complex and unique compared with those of their adult counterparts because they involve the interaction between normal growth and development and the optimization of athletic performance.^{37,38}

During the 1980s it was erroneously believed that leanness correlated with enhanced athletic performance as a result of studies that demonstrated a positive correlation between running performance and percentage of body fat.³⁹ Not only is scientific evidence lacking to prove that weight reduction alone improves athletic performance, but in fact deliberate caloric restriction in children and adolescents is likely to have detrimental implications, not only for their athletic performance but also for their growth and development and general health.³⁹ Unfortunately, these erroneous beliefs are often perpetuated nowadays by coaches who have little or usually no training in nutrition for athletes.³⁹ The employment of school-based coaches is often dependent on the success of their teams, and controlling an athlete's weight is often the easiest parameter by which a coach can try to ensure athletic success.³⁹ In fact, by reducing fat in the diet, it is possible that essential sources of protein and minerals and vitamins such as calcium, magnesium iron, zinc, B₁₂, and other fat-soluble vitamins that are critical for growth also may be eliminated from the diet.³⁸

Diet should play an integral role in any comprehensive training program, with specific attention to energy requirements, including appropriate combinations of protein, carbohydrates, fat, vitamins, and minerals.³⁸ These requirements are often subject to large interindividual variation not only between sporting codes but often within a given sport.³⁸

Results of the YRBS study in the 1990s confirmed that children and adolescents involved in regular sporting activities not only

maintain healthier diets including greater amounts of fruit and vegetables but are often less concerned with caloric intake and energy balance.⁴ For young athletes, the energy requirements must be sufficient to ensure normal growth and development but must also provide the additional calories to account for physical training.³⁸ The recommendations for estimated energy requirements in young athletes set by the Food and Nutrition Board are based on age, height, weight, and physical activity classification.³⁸

Protein is an essential part of a young athlete's diet because it is required to build amino acids necessary for the growth and development of lean body mass and healthy bones but also as an alternative to carbohydrates as a source of energy.³⁸ Research is lacking regarding the recommended daily protein intake for young athletes.³⁸ Twelve percent to 15% of the dietary energy of adults should come from protein; however, the energy demands of children are greater, especially when they are involved in competitive, intensive training during periods of rapid growth.³⁸

Research shows that children and adolescents up to the age of 13 to 15 years have restricted glycolytic capacity, which calls into question the role of high-carbohydrate diets for younger children.³⁸ Regardless, nutritionists recommend that at least 50% of a young athlete's diet consist of carbohydrate because of the importance of this energy source during high-intensity training.³⁹ A significant amount of research is needed with regard to the optimal nutrition of the pediatric athlete.

PERFORMANCE-ENHANCING SUBSTANCES

The use of performance-enhancing substances is increasing among children and adolescents as a result of media exposure, the availability of so-called *natural supplements*, the absence of formal drug testing in schools, and the increasingly competitive nature of youth sports.⁴⁰ Pediatric athletes are at high risk because of increased susceptibility to societal pressures at a time where they are often dealing with complex developmental and psychosocial changes.

The term *ergogenic* is derived from the Greek "to make work" and refers to the inherent ability of many substances to enhance athletic power and/or endurance.⁴⁰ In many cases, the ergogenic effects of a substance are actually secondary to their intended use.⁴⁰ It is therefore essential that physicians dealing with athletes, especially those competing in high-level sports, have a working knowledge of substances that contain ergogenic properties, because inappropriate prescribing/counseling may result in the disqualification of an athlete from a competition.⁴¹

Anabolic-Androgenic Steroids

Although a wide range of performance-enhancing substances are available in the United States, anabolic-androgenic steroids are by far the most publicized and intensely studied. Anabolic-androgenic steroids are a synthetic analog of the male hormone testosterone, and their use in the pediatric athlete for both performance and enhancement of the physique has been documented in the medical literature for well over 20 years.⁴⁰ The use of androgenic steroids is widespread; an estimated 4%

to 12% of male adolescents and 0.5% to 2% of female adolescents used anabolic-androgenic steroids in the 1990s even though they were banned by almost every major athletic governing body.^{41,88}

As the name suggests, anabolic-androgenic steroids have both masculinizing and tissue-building effects, and thus when they are used in conjunction with adequate strength training and proper diet, they have the ability to increase muscle size and strength, enabling high-intensity workouts and possibly even a reduced recovery time after workouts.⁴⁰ As a result, athletes who are attracted to the substance tend to be those whose sport requires strength (such as weightlifters, throwers, and football players) or frequent, high-intensity workouts (such as swimmers and runners).⁴⁰

Research conducted by Kindlundh et al.⁹⁰ in 1999 demonstrated a significant correlation between the use of anabolic-androgenic steroids in adolescents and the abuse of other common drugs such as alcohol, tobacco, cannabis, and opioids.

Although the perceived performance-enhancing benefits appear high, the adverse effects of using anabolic-androgenic steroids are extensive and often irreversible.⁴⁰ In addition to the personality changes and psychological problems that are associated with steroid use, premature closure of epiphyseal plates with subsequent linear growth arrest, irreversible alopecia, gynecomastia, and acne and irreversible masculinization of secondary sexual characteristics in females are just a few of the more dramatic and often psychologically devastating adverse effects of anabolic-androgenic steroids.^{40,42}

Regulation of Performance-Enhancing Substances

Drug testing is both time consuming and expensive, making the widespread testing of young athletes virtually impossible.³² Nonetheless, many schools and youth organizations have implemented voluntary drug testing, which has a dual benefit of identifying and assisting athletes with abuse problems and reducing the peer pressure to use drugs.³⁰

With the introduction of the Dietary Supplement Health and Education Act in 1994, the role of the Food and Drug Administration in regulating "natural supplements" was eliminated.¹⁵ Since that time, "natural agents" such as creatinine, androstenedione, and dehydroepiandrosterone (DHEA) have been widely accessible via health stores and the internet.¹⁵ This accessibility results in an erroneous perception that these substances are "safe," even though the absence of regulatory control eliminates any legal requirement of manufacturers to declare all active ingredients and potential interactions and fully test their products for short- and long-term effects.³⁰

The use of performance-enhancing drugs among athletes of any age is unethical, unhealthy, and potentially life threatening.³² As physicians we have a responsibility to acquire and impart factual knowledge to young athletes who are contemplating the use of these substances. Although the effectiveness of using scare tactics that emphasize the negative effects of substance use has been questioned, a clear role exists for positive counseling with regard to healthy alternatives such as strength training and conditioning, nutrition, and skill acquisition through coaching and camps.³²

ARTHROSCOPY IN CHILDREN

The use of arthroscopy in the pediatric and adolescent population has dramatically expanded during the past decade as a result of increased youth participation in sport and the subsequent rise in sports-related injuries.⁶ With the advent of smaller, more sophisticated arthroscopic instruments during the past decade, the major obstacle to the application of arthroscopy in children was overcome.⁶ In fact, after extensive experience, Gross⁴³ noted that despite the difference in joint size, basic techniques of arthroscopy are largely the same in both children and adults. At present, arthroscopy is indicated in the management of shoulder, elbow, wrist, hip, knee, and ankle injuries within the pediatric population.⁶ Advantages of arthroscopy in this population include reduced postoperative morbidity, smaller incisions, more rapid return to activities, decreased inflammatory response, and improved visualization of joint structures.⁶

Shoulder injuries in the pediatric athlete include acute fractures, overuse injuries such as little league shoulder (Fig. 131.1), and shoulder instability (Fig. 131.2). Most major shoulder injuries requiring arthroscopy are related to instability and can be divided into two descriptive groups: traumatic anterior instability and multidirectional instability.^{4,6,38-41,43-51}

The incidence of elbow injuries continues to increase as a result of the growing popularity of youth sports. Many of the elbow injuries are repetitive, overuse-type injuries, such as osteochondritis dissecans (OCD), which is prevalent in athletes who participate in baseball, racket sports, and gymnastics.⁵² In fact, “little league elbow” is now an accepted term for a common overuse injury in young throwing athletes, with etiologies including a fragmented medial epicondyle (Fig. 131.3), OCD (Figs. 131.4 and 131.5), ulnar hypertrophy, and medial epicondylitis.^{49,52-54}



Fig. 131.1 An example of “little league shoulder,” which is associated with widening of the proximal humeral physis as a result of repetitive overuse.

Wrist arthroscopy is not a commonly practiced treatment modality among pediatric and adolescent patients because many injuries achieve successful healing nonoperatively and because of the restricted size of the joint space.⁵⁴ Kocher et al.⁵⁵ noted an increasing incidence of repetitive use-type injuries such as triangular fibrocartilage injuries (Fig. 131.6), and they believe that arthroscopy is indicated for débridement or determination

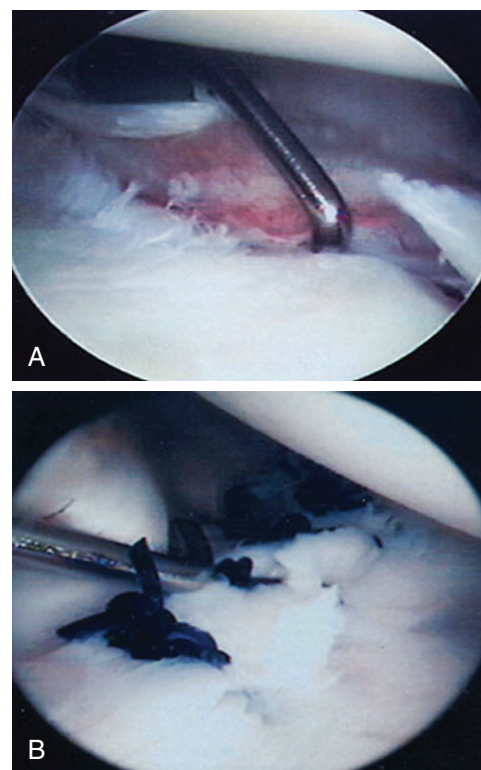


Fig. 131.2 Traumatic anterior shoulder instability. (A) Bankart lesion. (B) Repair of a Bankart lesion.



Fig. 131.3 Medial epicondyle widening associated with “little league elbow.”



Fig. 131.4 A sagittal magnetic resonance imaging scan of the elbow demonstrating a chondral defect of the capitulum associated with osteochondritis dissecans.



Fig. 131.5 A lateral radiograph of the elbow demonstrating a loose body in the anterior elbow.

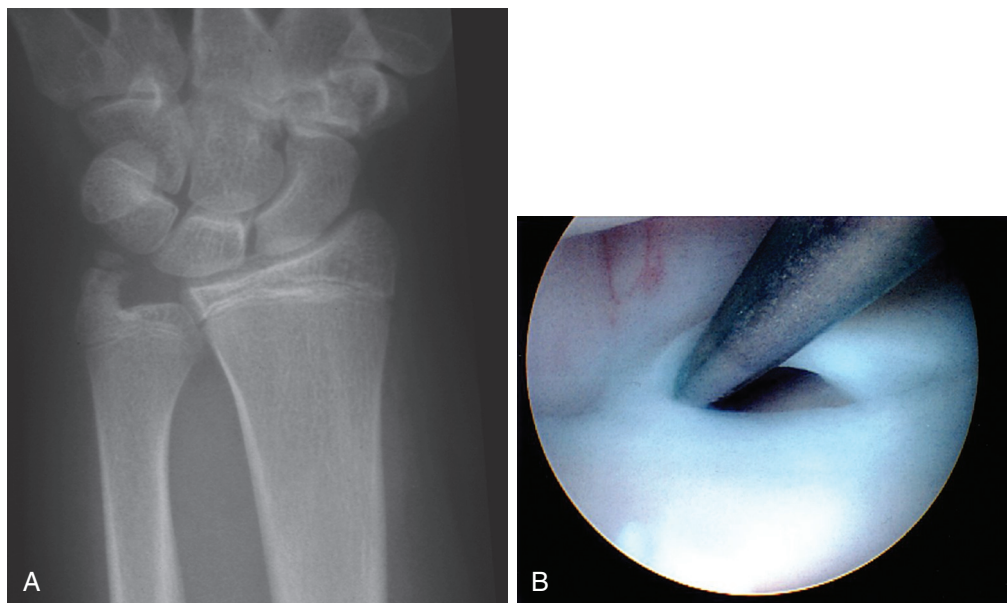


Fig. 131.6 An ulnar styloid fracture (A) associated with a triangular fibrocartilage tear (B).

of the extent of ligamentous injury in patients who fail to respond to nonoperative therapies.^{55,56}

Although hip arthroscopy is a commonly used diagnostic and treatment modality for pathologic conditions of the hip in the adult population, its application in the pediatric population is just beginning to increase. Indications in the pediatric population include isolated labral tears (Fig. 131.7), loose bodies, chondral injuries, and internal derangement associated with Perthes disease and epiphyseal dysplasias.⁵⁷⁻⁶⁰ The risk of complications, although small, does exist and includes pudendal nerve irritation and recurrent injury.⁶

Currently the largest application of arthroscopy in the pediatric and adolescent population is in the treatment of knee disease and is directly attributable to increased athletic activity.⁴³ Key indications for knee arthroscopy include OCD (Figs. 131.8 and 131.9), discoid meniscus, tibial spine fractures (Figs. 131.10 and 131.11), and partial and complete anterior cruciate ligament tears.⁶¹⁻⁶⁹

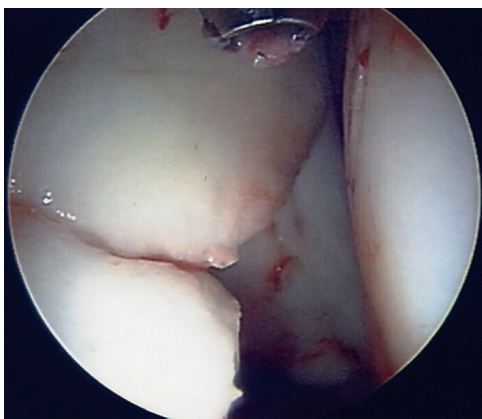


Fig. 131.7 A radial labral tear of the hip.

As more knee arthroscopy procedures are being performed on children and adolescents, additional investigations into the safety of such procedures have been performed. For example, venous thromboembolic events (VTEs), such as deep venous thrombosis (DVT) and pulmonary embolus (PE), which were previously thought to be exceedingly rare, have recently been reported in small series. One study from a high-volume pediatric hospital described a series of seven adolescent patients who developed symptomatic VTEs.⁸⁹ Interestingly, all patients had one or more risk factor, such as oral contraceptive use, smoking, obesity, or a concomitant open procedure at the time of arthroscopy, suggesting that perhaps select adolescents should undergo some form of DVT prophylaxis in the perioperative period. Given the rising volume of pediatric sports medicine surgeries, this remains an important area of future research to facilitate more definitive guidelines or recommendations.

Currently, the use of ankle arthroscopy in the pediatric population is restricted to a small number of conditions, including OCD, loose body removal, and triplane fracture repair, because of technical challenges resulting from the size of the joint and the risk of neurovascular damage.^{43,70-73}

CONCLUSIONS

Pediatric sports injuries are being seen with increased frequency. Just as the child is not a “little adult,” the pediatric athlete is not a “little adult athlete.” An understanding of the unique considerations of the pediatric athlete with respect to epidemiologic factors, endurance, flexibility, strength, thermoregulation, psychology, and nutrition is important background knowledge. Recognition of common injury patterns of the shoulder, elbow, wrist, hip, knee, and ankle is essential to effective management.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

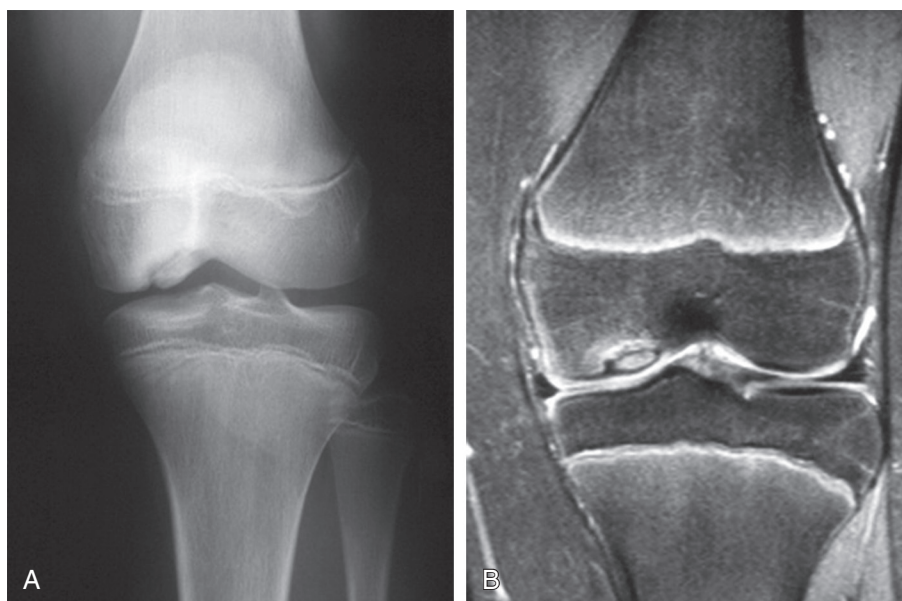


Fig. 131.8 Osteochondritis dissecans of the knee. (A) An anteroposterior radiograph. (B) A corresponding coronal magnetic resonance image.

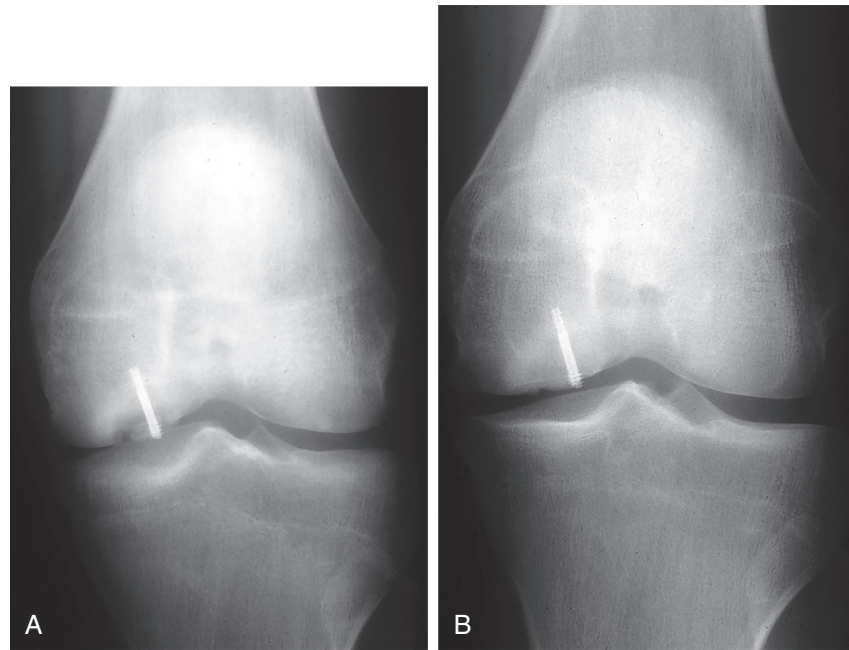


Fig. 131.9 Fixation of an unstable osteochondritis dissecans lesion of the knee. (A) An anteroposterior radiograph obtained immediately after the operation. (B) A radiograph obtained 3 months after the operation demonstrating healing of the lesion.

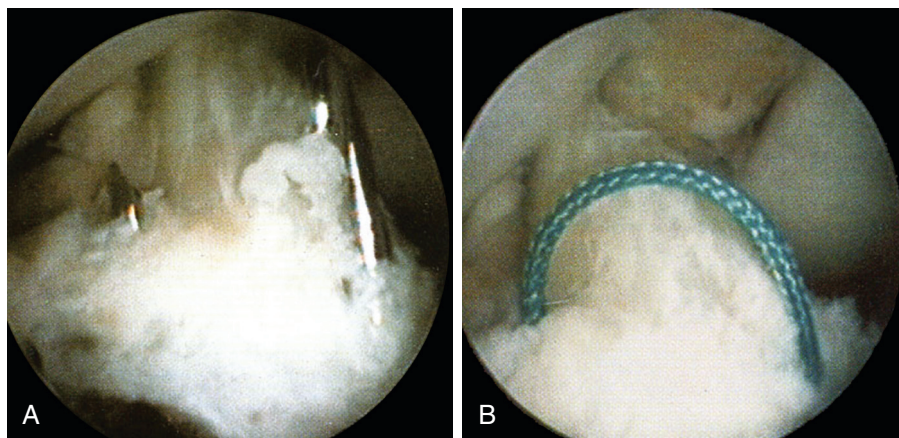


Fig. 131.10 Suture fixation of a tibial spine fracture. (A) Guidewires brought through the tibial spine fragment. (B) Suture fixation.

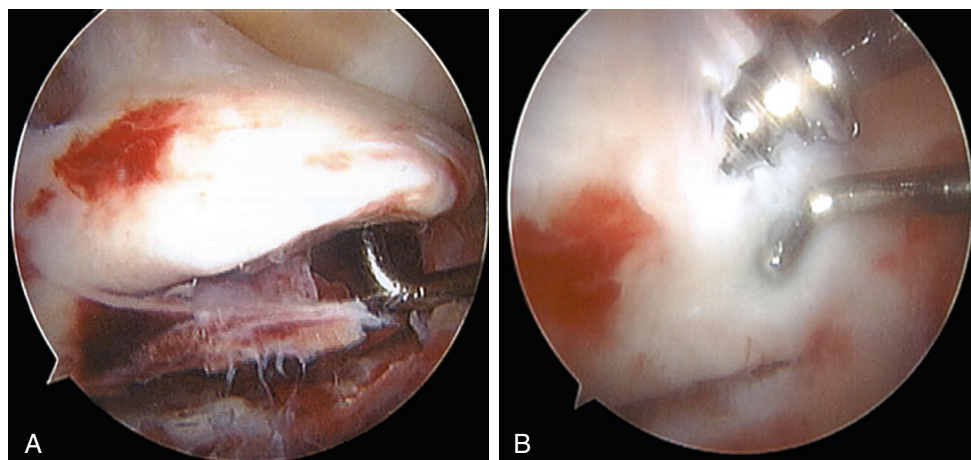


Fig. 131.11 Epiphyseal cannulated screw fixation of a tibial spine fracture. (A) A displaced fracture. (B) Screw fixation.

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